Multilevel Modeling for Longitudinal Data

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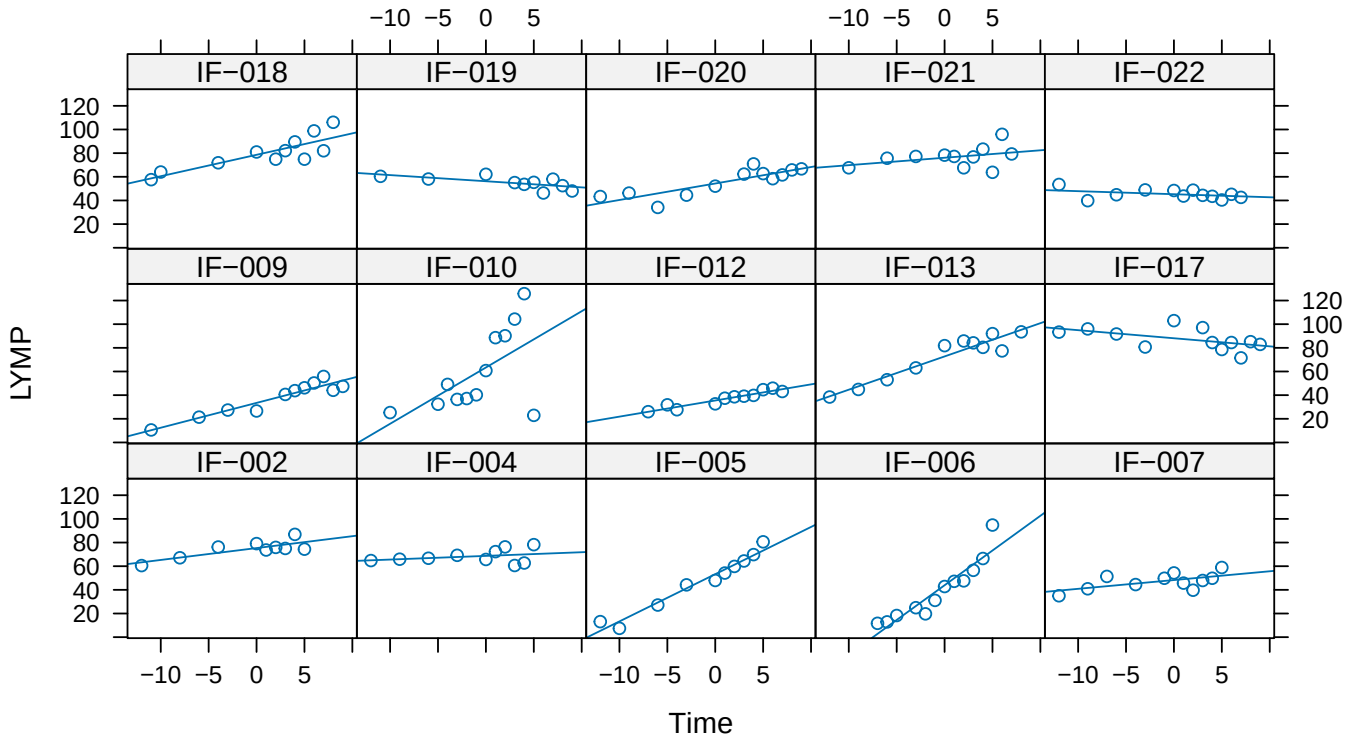
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Data description

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before the intervention, during the intervention, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] time point, with a frequency of every 30 days, treating the last pretest as Month 0;
- **Seq** - [numeric] time point within each treatment group, starting from Month 0;
- **Trt** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v2.csv", stringsAsFactors=T)
lattice::xyplot(LYMP ~ Time | ID, data=df, type=c("p","r"))
```



The hypothesis is that IF can help patients lower their lymphocyte count. The objectives are then to check whether (1) the means and (2) the linear slopes of time are statistically different before versus during the intervention.

Linear mixed-effect regression

Using the "nlme" package (v 3.1-162)

```
dat <- df[!is.na(df$LYMP) & df$Trt!=2,] #IF-013 contains one missing value

fit1 <- nlme::lme(LYMP ~ factor(Trt), random=~1 | ID, data=dat, method="ML")
summary(fit1)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1224.717 1236.706 -608.3586
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    15.96643 12.81501
##
## Fixed effects:  LYMP ~ factor(Trt)
##                 Value Std.Error  DF   t-value p-value
## (Intercept)  50.17287  4.422792 132 11.344163       0
## factor(Trt)1 14.12315  2.161386 132  6.534303       0
##  Correlation:
##              (Intr)
## factor(Trt)1 -0.246
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -2.49165610 -0.55618518 -0.03192352  0.62292969  4.25175278
##
## Number of Observations: 148
## Number of Groups: 15
```

```r
fit2 <- nlme::lme(LYMP ~ factor(Trt) * Time, random=~Time | ID, data=dat, method="ML")
summary(fit2)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##       AIC      BIC    logLik
##   1124.47 1148.447 -554.2349
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.522600 (Intr)
## Time         2.019187 -0.089
## Residual     7.148936
##
## Fixed effects:  LYMP ~ factor(Trt) * Time
##                       Value Std.Error  DF   t-value p-value
## (Intercept)       57.89561  4.558001 130 12.701974  0.0000
## factor(Trt)1       3.71169  2.519625 130  1.473110  0.1431
## Time               1.49179  0.573226 130  2.602454  0.0103
## factor(Trt)1:Time -0.13552  0.579038 130 -0.234052  0.8153
##  Correlation:
##                   (Intr) fc(T)1 Time
## factor(Trt)1      -0.177
## Time               0.019 -0.178
## factor(Trt)1:Time -0.100 -0.514 -0.144
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -3.26424016 -0.35417230 -0.04107843  0.49755880  3.70258899
##
## Number of Observations: 148
## Number of Groups: 15
```

Using the "lme4" package (v 1.1-33)

```r
fit1 <- lme4::lmer(LYMP ~ factor(Trt) + (1 | ID), dat, REML=F)
summary(fit1) #(same) output omitted

fit2 <- lme4::lmer(LYMP ~ factor(Trt) * Time + (Time | ID), dat, REML=F)
summary(fit2) #(same) output omitted
```

In linear mixed-effect regression (LMER), each subject has its intercept and slope (i.e., random intercepts and random slopes). The p-value of .8153 is not statistically significant at an alpha level of .05 ($t = -0.234052$, $df = 130$). Hence, fail to reject that the linear slopes of time are statistically different before versus during the intervention.

```
coef(fit2)
```

```
##        (Intercept) factor(Trt)1        Time factor(Trt)1:Time
## IF-002    73.87633     3.711686  0.93147491        -0.1355249
## IF-004    65.28458     3.711686 -0.09659144        -0.1355249
## IF-005    49.86903     3.711686  3.47626799        -0.1355249
## IF-006    39.00678     3.711686  4.29918136        -0.1355249
## IF-007    45.69031     3.711686  0.39434789        -0.1355249
## IF-009    32.89688     3.711686  2.02408234        -0.1355249
## IF-010    70.11684     3.711686  6.51099919        -0.1355249
## IF-012    33.75998     3.711686  0.98786257        -0.1355249
## IF-013    72.62868     3.711686  2.87679601        -0.1355249
## IF-017    86.11678     3.711686 -0.80831224        -0.1355249
## IF-018    74.08746     3.711686  1.16819162        -0.1355249
## IF-019    54.58306     3.711686 -0.63209714        -0.1355249
## IF-020    52.89076     3.711686  1.21778064        -0.1355249
## IF-021    73.98122     3.711686  0.43660057        -0.1355249
## IF-022    43.64540     3.711686 -0.40968697        -0.1355249
```

Check (3) any difference in either autophagy or lymphocyte count between those who fasted three versus six months.

```
dat2 <- df[!is.na(df$ATG) & df$Trt==1,] #Excluded IF-007 and 012

fit3 <- nlme::lme(ATG ~ Seq, random=~Seq | ID, data=dat2)
summary(fit3)
```

```
## Linear mixed-effects model fit by REML
##   Data: dat2
##        AIC      BIC   logLik
##   5.821146 18.28637 3.089427
##
## Random effects:
##  Formula: ~Seq | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev     Corr
## (Intercept) 0.31568064 (Intr)
## Seq         0.05766067 -0.853
## Residual    0.15799595
##
## Fixed effects:  ATG ~ Seq
##                 Value  Std.Error DF   t-value p-value
## (Intercept)  0.6337515 0.09426893 47  6.722804  0.0000
## Seq         -0.0536462 0.02129648 47 -2.519015  0.0152
##  Correlation:
##     (Intr)
## Seq -0.787
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -1.4623278 -0.5742309 -0.2324513  0.4079760  2.6236637
##
```

```
## Number of Observations: 61
## Number of Groups: 13
```